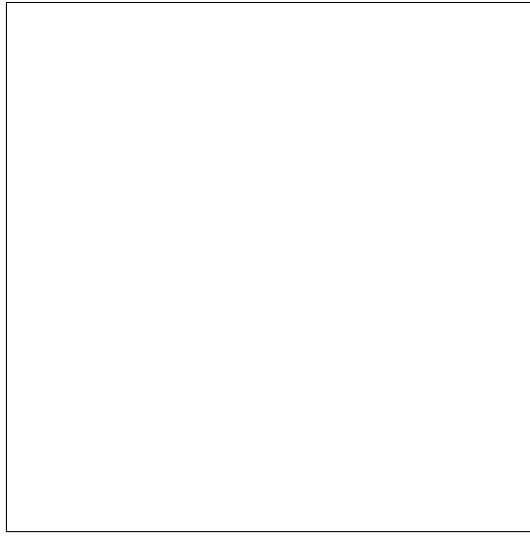




- Introduction. Focus on simulation of a mid-ocean ridge.
 - Model formulation. Give symbols and definitions of parameters
 - Physics model
 - * Flow. (Citing the zero porosity paper).
 - * Transport.
 - * Eutectic two phase system.
 - Numerical model
 - * MFEM and FVM scheme and coupling stratgy. (leave out the discretized form).
 - * Flow and thermal IC and BCs.
 - Numerical results, analysis and comparison with other models.
 - At steady state
 - * Primary variables distribution: Temperature, Velocity, pressure, and opx concentration.
 - * Velocity streamline with porosity presents. (Verification of mass conservation?)
 - Evolve with time
 - * Melt how it transported as time evolving. Try to explain melt focusing with our model.
 - * Porosity evolution and formation of porosity waves.
 - How new eutectic phase relation affected the result. Comparing with other phase transition model results.
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- Physics of the Earth and Planetary Interiors.
 - Geochemistry, Geophysics, Geosystems.